# Sagar Gupta

ProServe (Cloud Consultant) - DevOps/MLOps | Amazon Web Services (AWS) | Hyderabad, India

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## Summary

Cloud Consultant at AWS Professional Services with two years architecting multi-account AWS platforms and greenfield migrations for enterprise financial services clients. Currently building the Terraform foundation for RWS's data-centre exit and a second landing zone: Organizations OUs with Control Tower baselines, SCPs, tag policies, and two Transit Gateway hubs with Network Firewall inspection and VPN/Direct Connect. Worked on CCMv4.0-aligned landing zone controls at State Street (1,000+ preventative controls, team result) and modernized cloud-native IaC at DTCC. Terraform module and Landing Zone specialist; 11 upstream PRs merged (axios, Airflow, lucide, Prefect, Feast, awslabs/mcp, multer, and others). 5x AWS certified including Solutions Architect – Associate.

## Experience

**Amazon Web Services (AWS)** Aug 2024 – Present
*ProServe (Cloud Consultant) - DevOps/MLOps* Hyderabad, India

- **Lead DevOps Consultant – RWS (Feb 2026 – Present):** Built the AWS foundation for a data-centre exit and a second landing zone entirely in Terraform: Organizations OUs with Control Tower baselines, SCPs, tag policies, and a security baseline in every account (IMDSv2, EBS encryption, no default VPC). Delivered two Transit Gateway hubs in separate AWS Regions with Network Firewall inspection and VPN/Direct Connect, plus self-service onboarding: each new workload gets its route table within 15 minutes and firewall access by tag in about a minute. Won approval for AWS IPAM by measuring real billing in a sandbox (prefix delegation, about 94% lower projected cost). Cut per-account setup from 3+ days to under 4 hours (~90%).
- **DevOps Consultant – State Street (Oct 2024 – Aug 2025):** Built Terraform for about 10 AWS services with automated tests (tftest + PyTest) on the regulated landing zone team, part of the path that opened 38+ services to application teams. Designed Amazon Detective enablement with delegated admin in the Audit account via an AFT customization. Worked on the security solution behind 1,000+ preventative CCMv4.0 controls (team result). Project closed with a 10/10 CSAT.
- **DevOps Consultant – DTCC (Aug 2025 – Nov 2025):** Modernized legacy Terraform to cloud-native IaC patterns on high-sensitivity financial workloads. Used Amazon Q to accelerate code analysis and migration assessments across multiple workstreams.
- **MLOps Pipeline (SME Program, Jun – Aug 2025):** Provisioned end-to-end SageMaker MLOps infrastructure via Terraform + CodePipeline – blue-green endpoints with auto-scaling, CloudFront frontend, API Gateway + Lambda inference, drift-triggered retraining, and alarm-driven automatic rollback. Generalized and published by AWS as an official open-source sample (aws-samples, MIT-0).
- **Internal Impact:** Published a second AWS sample, Terraform for AWS Organizations governance on Control Tower (aws-samples, MIT-0), covering six policy types, Control Tower controls, and delegated admin for 15 services. Published APG pattern "Secure PR Previews and Approval-Gated Releases for Amplify Gen 2 with GitHub" on AWS Prescriptive Guidance (plus 2 more peer-reviewed in publication pipeline). MLOps SME Program Graduate. 5x TFC Ambassador (AI/ML, NGDE Graduated, CloudOps, Agentic AI, Serverless). 2 org-wide tech talks on Terraform testing and GitOps gating. Associate Speaker Certified.

**Amazon Web Services (AWS)** Mar 2024 – Jul 2024
*Associate Professional Services – DevOps (Internship)* Hyderabad, India

- Migrated on-premises application to AWS ECS with blue-green deployment via Terraform, CodePipeline, and CodeDeploy – zero-downtime releases with X-Ray and CloudWatch distributed tracing.

**Ikarus-3D** May 2023 – Jul 2023
*Software Developer Associate (Internship)* Mohali, India

- Containerized production application on AWS ECS/ECR with GitHub Actions CI/CD; deployed SonarQube on EC2 + RDS for continuous code quality analysis.

## Industry Certifications (5)

**AWS**: Solutions Architect – Associate | Developer – Associate | Machine Learning Engineer – Associate
**AWS**: AI Practitioner | Cloud Practitioner

## Education

**National Institute of Technology Warangal, Telangana** Sep 2021 – Jul 2024
Master of Computer Applications (MCA) | NIMCET AIR 208 CGPA: 8.38

**Sanskar College of Professional Studies (DAVV University), Indore** May 2018 – Aug 2021
Bachelor of Computer Applications (BCA) CGPA: 7.34

## Projects

**Blue-Green AWS Terraform** Jul 2025
*Terraform, ECS on EC2, ALB, CodePipeline, KMS, IAM* GitHub
◇ 15-module Terraform stack for zero-downtime blue-green deployments on ECS backed by an EC2 Auto Scaling Group – KMS encryption, ALB traffic shifting, and least-privilege IAM. Multi-account ready with separated networking, compute, and deployment layers; full CodePipeline automation for container builds and progressive traffic cutover.

**Ledger Sync** Jan 2026
*TypeScript, React, FastAPI, SQLite* Live | GitHub
◇ Personal-finance transaction reconciler with fuzzy-match dedup across bank statements and card exports; typed FastAPI backend, local-first SQLite store, React UI.

**GitScope** Mar 2026
*JavaScript, Chrome Manifest V3, GitHub GraphQL, Cloudflare Workers* Chrome Web Store | GitHub
◇ Chrome extension delivering GitHub repo insights via GraphQL API with a Cloudflare Worker backend for server-side OAuth – edge-compute architecture with 5-minute response caching and zero runtime dependencies.

**Claude Cost Optimizer** 2026
*React, Analytics* Live
◇ Web tool that models Claude API usage patterns and surfaces prompt-caching, batching, and model-tier changes worth 30–60% cost reduction on real workloads.

**Bedrock Multi-Model MCP** 2026
*Python, AWS Bedrock, MCP Protocol*
◇ MCP server routing a single client through 500+ Amazon Bedrock foundation models with unified auth and model-alias support – removes per-model SDK boilerplate for agent workflows.

**Other Notable Projects:** DevOps-AWS-FARM (Terraform + ECS + FastAPI/React/Mongo), LeetCode Rating Predictor (Python + TensorFlow, 121K+ contest records), Instagram Likes Leaderboard (TypeScript/Preact).

## Open Source Contributions

- **axios** (109K☆) – Synthesize AxiosError message from AggregateError on dual-stack connection failures (#11059, merged)
- **Apache Airflow** (44K☆) – Add template_fields to SalesforceBulkOperator (#63109, merged)
- **PrefectHQ/prefect** (21K☆) – Document custom deployment steps as Python functions (#20956, merged)
- **aws-ia/terraform-aws-vpc** – Root-caused the Cloud WAN attachment-replacement bug on `create_vpc = false`; fix co-authored into v4.8.0 (merged)
- **hashicorp/terraform-provider-aws** (10K☆) – Fix IoT cloudwatch_metric substitution template validation (#46867, open)
- **terraform-aws-modules/atlantis** – Pass custom egress rules through to the ECS service security group (#446, merged)
- **lucide** (24K☆) – Support kebab-case icon-node prop on the Vue Icon component (#4512, merged)
- **feast-dev/feast** (6K☆) – Claude Code agent skills for Feast (#6081, merged)
- **awslabs/agent-plugins** – defusedxml extension guard (#132, merged)
- **awslabs/mcp** – Fix Kendra documentation menu to match the source directory (#2607, merged)
- **expressjs/multer** – Stop files skipped by fileFilter from consuming maxCount (#1426, merged)

## Technical Skills

**AWS Multi-Account & Governance**: Organizations, OU Design, SCPs, Tag Policies, Landing Zone, Control Tower, AWS Config, Security Hub, IAM Identity Center, Service Catalog, CloudTrail, GuardDuty
**Networking & Hybrid Connectivity**: Transit Gateway (hub-and-spoke), VPC, VPN, Direct Connect, PrivateLink, Route 53 Resolver, Centralized Egress Inspection, Network Firewall
**Compute & Application Services**: ECS Fargate, Lambda, API Gateway, CloudFront, ALB/NLB, EventBridge, Step Functions, SageMaker
**Infrastructure as Code**: Terraform (modules, tftest, PyTest), CloudFormation, HCL, Multi-Account Strategy, Docker
**CI/CD & Automation**: GitHub Actions, CodePipeline, CodeBuild, CodeDeploy, Blue-Green Deployments, GitOps, Policy-as-Code
**Security & Compliance**: CCMv4.0, Preventative SCPs, KMS, Least-Privilege IAM, Config Rules, Security Hub Standards, Amazon Q
**Languages**: HCL, Python, TypeScript, JavaScript, Bash, SQL

## Achievements

- Published APG pattern "Secure PR Previews and Approval-Gated Releases for Amplify Gen 2 with GitHub" on AWS Prescriptive Guidance (plus 2 more peer-reviewed in publication pipeline). MLOps SME Program Graduate. Associate Speaker Certified.
- 5x TFC Ambassador – AI/ML, NGDE (Graduated), CloudOps (AWS DevOps Agent), Agentic AI, Serverless.
- Rank 1289 (Top 5%) at Google Kick Start Round E '22. *Certificate*
- 1st place – Coding Cauldron '23 (NIT Trichy); 1st place – QFiesta IGNOSI '22 (NIT Warangal).
- LeetCode Guardian badge – 2166 peak rating, 1300+ problems solved, 107 contests. 500+ DSA problems on GeeksforGeeks. HackerRank 6-Star Problem Solving, 5-Star C++.